

Capítulo 3

1. Which of the following data types can be used in a `switch` expression? (Choose all that apply.)

☒ A. `enum`
☐ B. `int`
☐ C. `Byte`
☐ D. `long`
☐ E. `String`
☐ F. `char`
☐ G. `var`
☐ H. `double`

2. What is the output of the following code snippet? (Choose all that apply.)

```
3: int temperature = 4;  
4: long humidity = -temperature + temperature * 3;  
5: if (temperature >= 4)  
6: if (humidity < 6) System.out.println("Too Low");  
7: else System.out.println("Just Right");  
8: else System.out.println("Too High");
```

☐ A. Too Low
☒ B. Just Right
☐ C. Too High
☐ D. A `NullPointerException` is thrown at runtime.
☐ E. The code will not compile because of line 7.
☐ F. The code will not compile because of line 8.

Humidity = 8, por lo que entra en el primer `if` (línea 5), al estar anidado y al incumplir la segunda sentencia (línea 6) el código termina en la línea 7

3. Which of the following data types are permitted on the right side of a for-each expression?
(Choose all that apply.)

- ☒ A. Double[][]
- ☐ B. Object
- ☐ C. Map
- ☒ D. List
- ☐ E. String
- ☒ F. char[]
- ☐ G. Exception
- ☒ H. Set

Un for-each admite arreglos, A y F son correctas, list y set es correcta porque hereda de collection – iterable que a su vez se pueden usar en el for-each.

4. What is the output of calling printReptile(6)?

```
void printReptile(int category) {  
    var type = switch(category) {  
        case 1,2 -> "Snake";  
        case 3,4 -> "Lizard";  
        case 5,6 -> "Turtle";  
        case 7,8 -> "Alligator";  
    };  
    System.out.print(type);  
}
```

- ☐ A. Snake
- ☐ B. Lizard
- ☐ C. Turtle
- ☐ D. Alligator
- ☐ E. TurtleAlligator
- ☒ F. None of the above

Es necesario agregar un default para que compile correctamente de no ser así el código no tiene un límite declarado por lo que mandaría un error

5. What is the output of the following code snippet?

```
List<Integer> myFavoriteNumbers = new ArrayList<>();  
myFavoriteNumbers.add(10);  
myFavoriteNumbers.add(14);  
for (var a : myFavoriteNumbers) {  
    System.out.print(a + ", ");  
    break;  
}  
  
for (int b : myFavoriteNumbers) {  
    continue;  
    System.out.print(b + ", ");  
}  
  
for (Object c : myFavoriteNumbers)  
    System.out.print(c + ", ");
```

- A. It compiles and runs without issue but does not produce any output.
- B. 10, 14,
- C. 10, 10, 14,
- D. 10, 10, 14, 10, 14,
- E. Exactly one line of code does not compile.**
- F. Exactly two lines of code do not compile.
- G. Three or more lines of code do not compile.
- H. The code contains an infinite loop and does not terminate.

En el segundo for aparece un continue dentro del mismo, al no ser accesible en el código marca un error.

6. Which statements about decision structures are true? (Choose all that apply.)

- A. A for-each loop can be executed on any Collections Framework object.
- B. The body of a while loop is guaranteed to be executed at least once.
- C. The conditional expression of a for loop is evaluated before the first execution of the loop body.**
- D. A switch expression that takes a String and assigns the result to a variable requires a default branch.**
- E. The body of a do/while loop is guaranteed to be executed at least once.**
- F. An if statement can have multiple corresponding else statements.

El cuerpo de un do/while se ejecuta 0 o mas veces por lo que E es correcta, la condición se evalua al inicio del bucle por lo que C es correcta y por ultimo forzosamente un switch requiere de una rama predeterminada por lo que D es correcta

7. Assuming `weather` is a well-formed nonempty array, which code snippet, when inserted independently into the blank in the following code, prints all of the elements of `weather`? (Choose all that apply.)

```
private void print(int[] weather) {  
    for(_____) {  
        System.out.println(weather[i]);  
    }  
}
```

- A. `int i=weather.length; i>0; i--`
- B. `int i=0; i<=weather.length-1; ++i`
- C. `var w : weather`
- D. `int i=weather.length-1; i>=0; i--`
- E. `int i=0, int j=3; i<weather.length; ++i`
- F. `int i=0; ++i<10 && i<weather.length;`
- G. None of the above

Es una forma correcta de ejecutar e imprimir el código, cumple con los requisitos de un `for` y su estructura.

8. What is the output of calling `printType(11)`?

```
31: void printType(Object o) {  
32:     if(o instanceof Integer bat) {  
33:         System.out.print("int");  
34:     } else if(o instanceof Integer bat && bat < 10) {  
35:         System.out.print("small int");  
36:     } else if(o instanceof Long bat || bat <= 20) {  
37:         System.out.print("long");  
38:     } default {  
39:         System.out.print("unknown");  
40:     }  
41: }
```

- A. `int`
- B. `small int`
- C. `long`
- D. `unknown`
- E. Nothing is printed.
- F. The code contains one line that does not compile.
- G. The code contains two lines that do not compile.
- H. None of the above

El uso de default no es requerido dentro de un for por lo cual es error de sintaxis (línea 38), el error en la línea 36 es un error del uso de bat dentro de un operador, el lado derecho (bat <= 20) solo se va a evaluar si el lado izquierdo es false, al evaluarlo si el primer bloque falla se llama a una variable que no existe en el compilador

9. Which statements, when inserted independently into the following blank, will cause the code to print 2 at runtime? (Choose all that apply.)

```
int count = 0;
BUNNY: for(int row = 1; row <=3; row++)
    RABBIT: for(int col = 0; col <3 ; col++) {
        if((col + row) % 2 == 0)
            _____;
        count++;
    }
System.out.println(count);
```

- A. break BUNNY
- B. break RABBIT**
- C. continue BUNNY**
- D. continue RABBIT
- E. break**
- F. continue
- G. None of the above, as the code contains a compiler error.

Cualquiera de las 3 opciones seleccionadas da un flujo en el código

10. Given the following method, how many lines contain compilation errors? (Choose all that apply.)

```
10: private DayOfWeek getWeekDay(int day, final int thursday) {
11:     int otherDay = day;
12:     int Sunday = 0;
13:     switch(otherDay) {
14:         default:
15:             case 1: continue;
16:             case thursday: return DayOfWeek.THURSDAY;
17:             case 2,10: break;
```

```

18:         case Sunday: return DayOfWeek.SUNDAY;
19:         case DayOfWeek.MONDAY: return DayOfWeek.MONDAY;
20:     }
21:     return DayOfWeek.FRIDAY;
22: }

```

- A. None, the code compiles without issue.
- B. 1
- C. 2
- D. 3
- E. 4**
- F. 5
- G. 6
- H. The code compiles but may produce an error at runtime.

Continue no se usa dentro de un switch, en la línea 16 al no ser una constante puede admitir cualquier valor alterando el comportamiento del código, falta final en la línea 18 y la línea 19 no compila por inconsistencia en el tipo de dato, no retorna un int.

11. What is the output of calling `printLocation(Animal.MAMMAL)?`

```

10: class Zoo {
11:     enum Animal {BIRD, FISH, MAMMAL}
12:     void printLocation(Animal a) {
13:         long type = switch(a) {
14:             case BIRD -> 1;
15:             case FISH -> 2;
16:             case MAMMAL -> 3;
17:             default -> 4;
18:         };
19:         System.out.print(type);
20:     } }

```

- A. 3**
- B. 4
- C. 34
- D. The code does not compile because of line 13.
- E. The code does not compile because of line 17.
- F. None of the above

12. What is the result of the following code snippet?

```
3: int sing = 8, squawk = 2, notes = 0;
4: while(sing > squawk) {
5:     sing--;
6:     squawk += 2;
```

```
7:     notes += sing + squawk;
8: }
9: System.out.println(notes);
```

- A. 11
- B. 13
- C. 23
- D. 33
- E. 50
- F. The code will not compile because of line 7.

Vuelta 1: notes = 11

Vuelta 2: notes = 23

Vuelta 3: bucle termina. Dando como respuesta 23

13. What is the output of the following code snippet?

```
2: boolean keepGoing = true;
3: int result = 15, meters = 10;
4: do {
5:     meters--;
6:     if(meters==8) keepGoing = false;
7:     result -= 2;
8: } while keepGoing;
9: System.out.println(result);
```

- A. 7
- B. 9
- C. 10
- D. 11
- E. 15
- F. The code will not compile because of line 6.
- G. The code does not compile for a different reason.

Faltan los paréntesis en la línea 8

14. Which statements about the following code snippet are correct? (Choose all that apply.)

```
for(var penguin : new int[2])
    System.out.println(penguin);
var ostrich = new Character[3];
for(var emu : ostrich)
    System.out.println(emu);
List<Integer> parrots = new ArrayList<Integer>();
for(var macaw : parrots)
    System.out.println(macaw);
```

- A. The data type of penguin is Integer.
- B. The data type of penguin is int.**
- C. The data type of emu is undefined.
- D. The data type of emu is Character.**
- E. The data type of macaw is List.
- F. The data type of macaw is Integer.
- G. None of the above, as the code does not compile.

Cada elemento de penguin es correcto tener un int por lo que la opción B es correcta, es correcto admitir Character en un for por lo que también es correcto

15. What is the result of the following code snippet?

```
final char a = 'A', e = 'E';
char grade = 'B';
switch (grade) {
    default:
    case a:
    case 'B': 'C': System.out.print("great ");
    case 'D': System.out.print("good "); break;
    case e:
    case 'F': System.out.print("not good ");
}
```

- A. great
- B. great good
- C. good
- D. not good
- E. The code does not compile because the data type of one or more case statements does not match the data type of the switch variable.
- F. None of the above**

Tiene sintaxis invalida, hacen falta : en el segundo case

16. Given the following array, which code snippets print the elements in reverse order from how they are declared? (Choose all that apply.)

```
char[] wolf = {'W', 'e', 'b', 'b', 'y'};
```

A.

```
int q = wolf.length;
for( ; ; ) {
    System.out.print(wolf[--q]);
    if(q==0) break;
}
```

B.

```
for(int m=wolf.length-1; m>=0; --m)
    System.out.print(wolf[m]);
```

C.

```
for(int z=0; z<wolf.length; z++)
    System.out.print(wolf[wolf.length-z]);
```

D.

```
int x = wolf.length-1;
for(int j=0; x>=0 && j==0; x--)
    System.out.print(wolf[x]);
```

E.

```
final int r = wolf.length;
for(int w = r-1; r>-1; w = r-1)
    System.out.print(wolf[w]);
```

F.

```
for(int i=wolf.length; i>0; --i)
    System.out.print(wolf[i]);
```

G. None of the above

En A cumple con declarar el for, pre-decremento y rompe el ciclo con el break, por lo tanto funciona.

En B Correcto uso de length y decremento valido, por lo que es correcto

En D Su decremento es correcto, como j nunca cambia de valor siempre es verdadero, por lo que la condición depende totalmente de $x \geq 0$.

17. What distinct numbers are printed when the following method is executed? (Choose all that apply.)

```
private void countAttendees() {  
    int participants = 4, animals = 2, performers = -1;  
    while((participants = participants+1) < 10) {}  
    do {} while (animals++ <= 1);  
    for( ; performers<2; performers+=2) {}  
  
    System.out.println(participants);  
    System.out.println(animals);  
    System.out.println(performers);  
}
```

- A. 6
- B. 3**
- C. 4
- D. 5
- E. 10**
- F. 9
- G. The code does not compile.
- H. None of the above

El primer bucle termina cuando llegue a 10 por lo que la opción G y H son incorrectas, el segundo bucle inicializa con un valor distinto de 1 alcanzando 3.

18. Which statements about pattern matching and flow scoping are correct? (Choose all that apply.)

- A. Pattern matching with an `if` statement is implemented using the `instance` operator.
- B. Pattern matching with an `if` statement is implemented using the `instanceon` operator.
- C. Pattern matching with an `if` statement is implemented using the `instanceof` operator.**
- D. The pattern variable cannot be accessed after the `if` statement in which it is declared.
- E. Flow scoping means a pattern variable is only accessible if the compiler can discern its type.**
- F. Pattern matching can be used to declare a variable with an `else` statement.

19. What is the output of the following code snippet?

```
2: double iguana = 0;
3: do {
4:     int snake = 1;
5:     System.out.print(snake++ + " ");
6:     iguana--;
7: } while (snake <= 5);
8: System.out.println(iguana);
```

- A. 1 2 3 4 -4.0
- B. 1 2 3 4 -5.0
- C. 1 2 3 4 5 -4.0
- D. 0 1 2 3 4 5 -5.0
- E. The code does not compile.**
- F. The code compiles but produces an infinite loop at runtime.
- G. None of the above

Snake esta fuera del alcance del resto del programa por lo que marca error de compilación

20. Which statements, when inserted into the following blanks, allow the code to compile and run without entering an infinite loop? (Choose all that apply.)

```
4: int height = 1;
5: L1: while(height++ <10) {
6:     long humidity = 12;
7:     L2: do {
8:         if(humidity-- % 12 == 0) _____;
9:         int temperature = 30;
10:        L3: for( ; ; ) {
11:            temperature++;
12:            if(temperature>50) _____;
13:        }
14:    } while (humidity > 4);
15: }
```

- A. break L2 on line 8; continue L2 on line 12
- B. continue on line 8; continue on line 12
- C. break L3 on line 8; break L1 on line 12
- D. continue L2 on line 8; continue L3 on line 12
- E. continue L2 on line 8; continue L2 on line 12
- F. None of the above, as the code contains a compiler error

La opción A es correcta ya que saca el bucle antes de adentrarse al bucle mas profundo y caer en un bucle infinito

21. A minimum of how many lines need to be corrected before the following method will compile?

```
21: void findZookeeper(Long id) {  
22:     System.out.print(switch(id) {  
23:         case 10 -> {"Jane"}  
24:         case 20 -> {yield "Lisa";};  
25:         case 30 -> "Kelly";  
26:         case 30 -> "Sarah";  
27:         default -> "Unassigned";  
28:     });  
29: }
```

- A. Zero
- B. One
- C. Two
- D. Three
- E. Four
- F. Five

Un long no es admitido en un switch, hace falta un punto y coma en la línea 23 y las líneas 25 y 26 tienen el mismo valor de case

22. What is the output of the following code snippet? (Choose all that apply.)

```
2: var tailFeathers = 3;
3: final var one = 1;
4: switch (tailFeathers) {
5:     case one: System.out.print(3 + " ");
6:     default: case 3: System.out.print(5 + " ");
7: }
8: while (tailFeathers > 1) {
9:     System.out.print(--tailFeathers + " "); }
```

- A. 3
- B. 5 1
- C. 5 2
- D. 3 5 1
- E. 5 2 1**
- F. The code will not compile because of lines 3–5.
- G. The code will not compile because of line 6.

Var es compatible con switch y while por lo que ejecuta sin problema, el bucle while se ejecuta dos veces inicializando en 3 y terminando en 2 y 1, el bucle switch da como salida 5.

23. What is the output of the following code snippet?

```
15: int penguin = 50, turtle = 75;
16: boolean older = penguin >= turtle;
17: if (older = true) System.out.println("Success");
18: else System.out.println("Failure");
19: else if(penguin != 50) System.out.println("Other");
```

- A. Success
- B. Failure
- C. Other
- D. The code will not compile because of line 17.
- E. The code compiles but throws an exception at runtime.
- F. None of the above**

Se inicializa un else sin haber tenido un if previamente

24. Which of the following are possible data types for friends that would allow the code to compile? (Choose all that apply.)

```
for(var friend in friends) {  
    System.out.println(friend);  
}
```

- A. Set
- B. Map
- C. String
- D. int[]
- E. Collection
- F. StringBuilder
- G. None of the above

No es un for valido ya que utiliza una palabra clave inexistente

25. What is the output of the following code snippet?

```
6: String instrument = "violin";  
7: final String CELLO = "cello";  
8: String viola = "viola";  
9: int p = -1;  
10: switch(instrument) {  
11:     case "bass" : break;  
12:     case CELLO : p++;  
13:     default: p++;  
14:     case "VIOLIN": p++;  
15:     case "viola" : ++p; break;  
16: }  
17: System.out.print(p);
```

- A. -1
- B. 0
- C. 1
- D. 2
- E. 3
- F. The code does not compile.

Incrementa 3 veces por lo que da valor final en 2

26. What is the output of the following code snippet? (Choose all that apply.)

```
9: int w = 0, r = 1;
10: String name = "";
11: while(w < 2) {
12:     name += "A";
13:     do {
14:         name += "B";
15:         if(name.length()>0) name += "C";
16:         else break;
17:     } while (r <=1);
18:     r++; w++; }
19: System.out.println(name);
```

- A. ABC
- B. ABCABC
- C. ABCABCABC
- D. Line 15 contains a compilation error.
- E. Line 18 contains a compilation error.
- F. The code compiles but never terminates at runtime.**
- G. The code compiles but throws a `NullPointerException` at runtime.

R al actualizarse fuera del bucle, genera que internamente nunca rompa el ciclo, generando un bucle infinito

27. What is printed by the following code snippet?

```
23: byte amphibian = 1;
24: String name = "Frog";
25: String color = switch(amphibian) {
26:     case 1 -> { yield "Red"; }
27:     case 2 -> { if(name.equals("Frog")) yield "Green"; }
28:     case 3 -> { yield "Purple"; }
29:     default -> throw new RuntimeException();
30: };
31: System.out.print(color);
```

- A. Red
- B. Green
- C. Purple
- D. RedPurple
- E. An exception is thrown at runtime.
- F. The code does not compile.**

En el case al no recibir ningún valor que sea igual a frog no compila por lo que la respuesta correcta es F

28. What is the output of calling `getFish("goldie")`?

```
40: void getFish(Object fish) {  
41:     if (!(fish instanceof String guppy))  
42:         System.out.print("Eat!");  
43:     else if (!(fish instanceof String guppy)) {  
44:         throw new RuntimeException();  
45:     }  
46:     System.out.print("Swim!");  
47: }
```

- A. Eat!
- B. Swim!
- C. Eat! followed by an exception.
- D. Eat!Swim!
- E. An exception is printed.
- F. None of the above**

En la línea 43 genera un duplicado de guppy declarado en la línea 41, debe de ser un nombre diferente para que el código pueda funcionar

29. What is the result of the following code?

```
1: public class PrintIntegers {  
2:     public static void main(String[] args) {  
3:         int y = -2;  
4:         do System.out.print(++y + " ");  
5:         while(y <= 5);  
6:     } }
```

- A. -2 -1 0 1 2 3 4 5
- B. -2 -1 0 1 2 3 4
- C. -1 0 1 2 3 4 5 6
- D. -1 0 1 2 3 4 5
- E. The code will not compile because of line 5.
- F. The code contains an infinite loop and does not terminate.

El primer valor es -1, en la penúltima iteración del bucle incrementa a 5, el bucle sigue su condición dando como ultimo valor 6